



Dynamic visualizations

Hey, great to see you again. So far we've shown that there's lots of choices you'll make as a data analyst when creating visualizations. Each of your choices should help make sure that your visuals are meaningful and effective. Another choice you'll need to make is whether you want your visualizations to be static or dynamic. Static visualizations do not change over time unless they're edited. They can be useful when you want to control your data and your data story. Any visualization printed on paper is automatically static.

Charts and graphs created in spreadsheets are often static too. For example, the owner of this spreadsheet might have to change the data in order for the visualization to update. Now, dynamic visualizations are interactive or change over time. The interactive nature of these graphics means that users have some control over what they see. This can be helpful if stakeholders want to adjust what they're able to view. Let's check out a visualization about happiness that we've created in Tableau. Tableau is a business intelligence and analytics platform that helps people see, understand, and make decisions with data.

Visualizations in Tableau are automatically interactive. We'll go into the dashboard to see how the happiness score has changed from 2015 to 2017. We can check this out in our 12th slide, yearly happiness changes. On the left are the country level changes in happiness score. The countries are sorted by largest increase to largest decrease. On the right, there's a map with overall happiness scores. The color scale moves from blue for the countries with the highest happiness score, to red for those with the lowest.

If you look below the map, you'll notice a year to view slider where people can choose which year's happiness scores to display on the map. It's currently set for 2016, but if someone wants to know the scores for 2015 or 2017, they can adjust the slider. They could then make note of how the color-coding and score labels change from year to year. Other dynamic visualizations upload new data automatically. These bar graphs continually update data by the minute and second. Other data visuals can do the same by day, week or month. If you need to, you can show trends in real-time.

Having an interactive visualization can be useful for both you and the audience you share it with. But it's good to remember that the more power you give the user, the less control you have over the story you want the data to tell. It's something to keep in mind as you learn how to create your own visualizations. You want to find the right balance between interactivity and control. Something else to consider is, a choice between using a static or dynamic visualization. This will usually depend on the data you're visualizing, the audience you're presenting to, and how you're giving your presentation. Now that we've made some decisions about what kind of data vis we want to create, we can start thinking about the design, which is exactly where we're going to start talking about next time. See you there.